배관 차단장치의 구조적 안정성 검증 용역 < 1650A >

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PART 1 연구 개요

◆ 연구 목표

- 배관 차단장치의 구조적 안정성 평가

◆ 연구 수행 내용

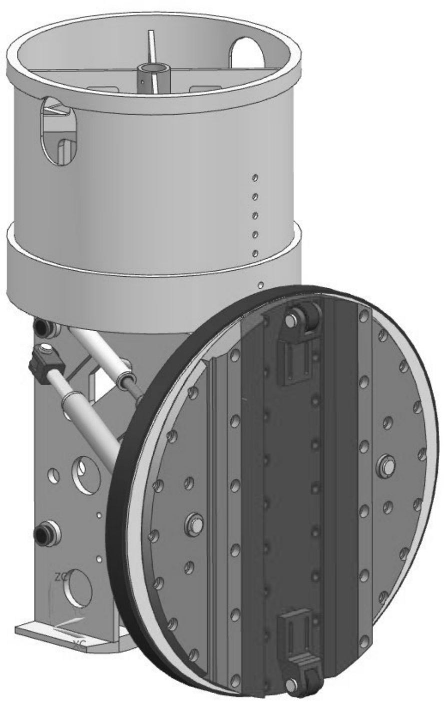
- 배관 차단장치 3D 모델링
  - 구조해석 진행 전, 모델링 상용 프로그램인 ANSYS Spaceclaim을 사용하여 배관 차단장치를 3차원 형상으로 구현하는 작업

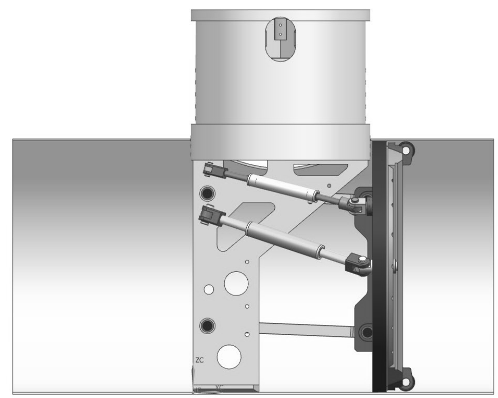
- 배관 차단장치 구조해석
  - 구조해석 상용 프로그램인 ANSYS Static Structural을 사용하여 하중에 따른 배관 차단장치의 변형 및 응력 계산

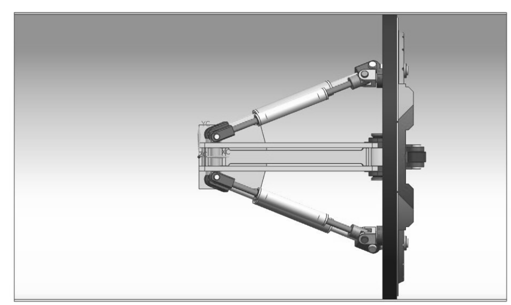
❖ 배관 차단장치 구조해석을 위한 형상 및 재료는 에서 제공한 도면(5%의 진원도)을 활용함.
❖ 본 해석은 각 요소(component)의 평균 응력을 기반으로 안정성 평가를 진행함.
❖ 본 해석의 평가는 서지압력(surge pressure) 또는 수격 현상(water hammer) 등의 이상 하중(unusual load)과 도면에 포함되지 않은 외적인 사항은 고려하지 않음.

◆ 해석 모델

- Model cases for Research
  - 1650A







◆ 해석 조건

- Partial Structural Analysis in Steady State
  - coupled with other parts for numerical efficiency and accuracy

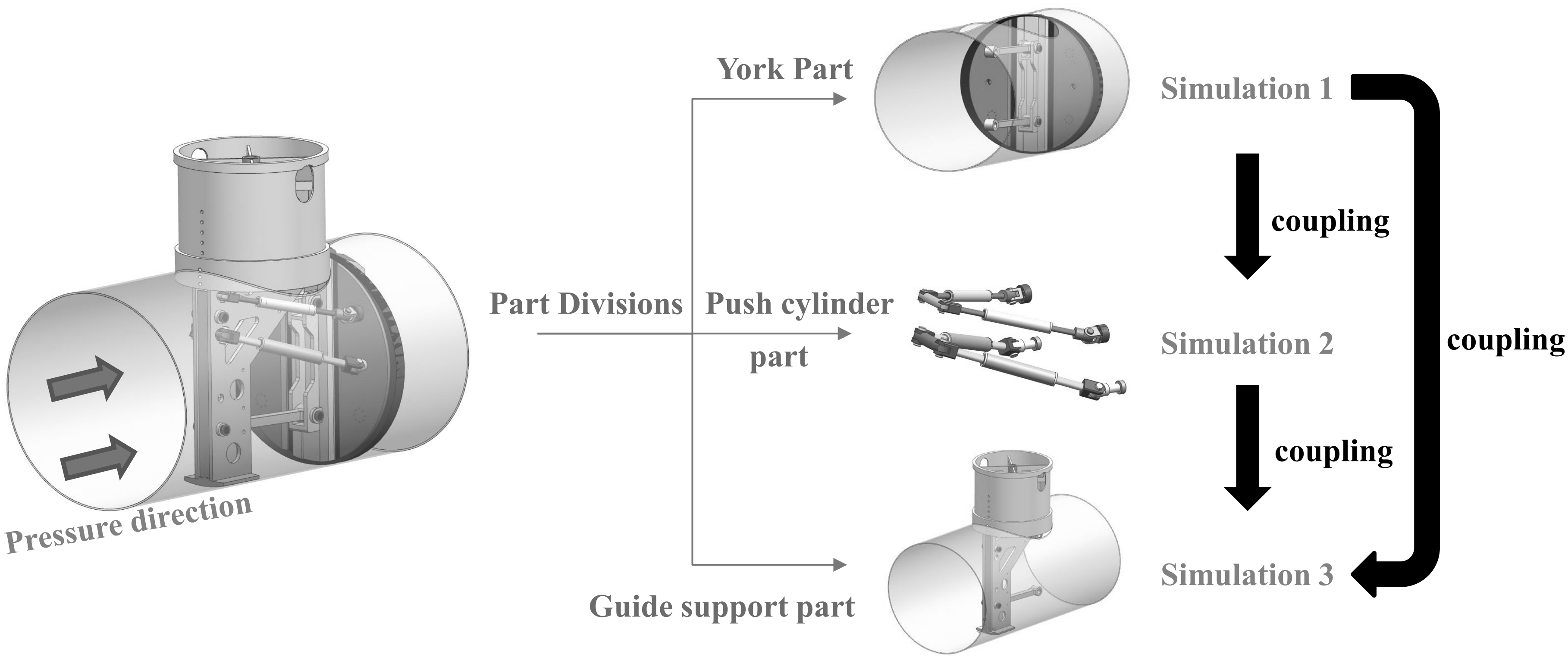


Fig. 1 Schematic diagram of numerical analysis procedure

◆ 해석 조건

- York part
  - Components

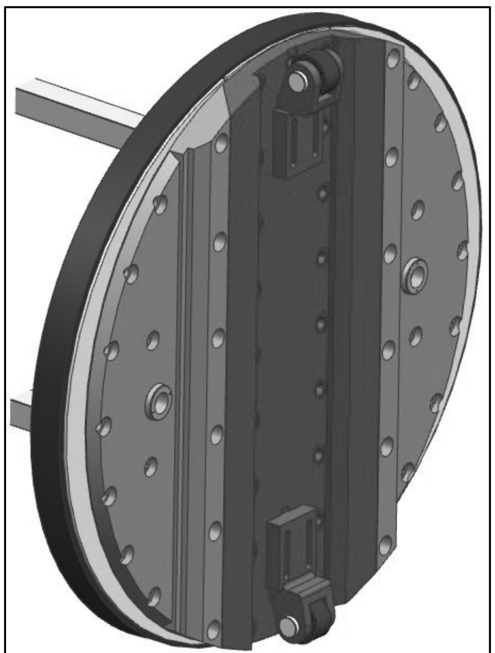


Fig. 2 York part

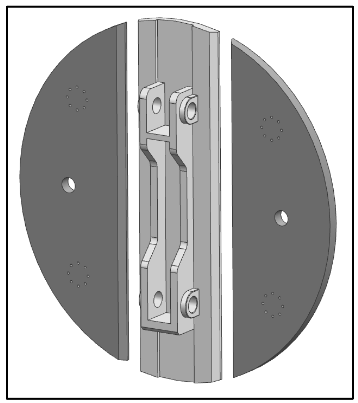


Fig. 3 Element plate

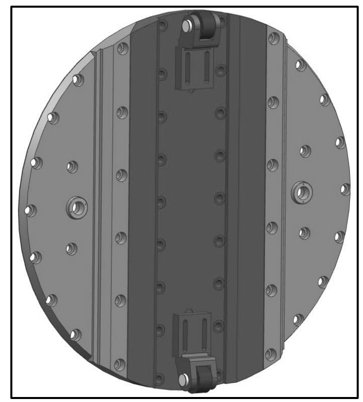


Fig. 4 York & Roller

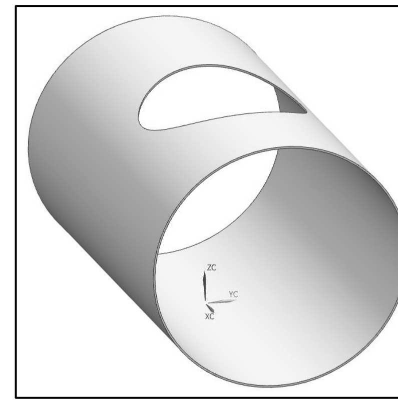


Fig. 5 Pipe

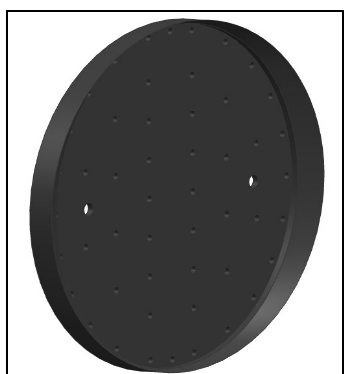


Fig. 6 Element

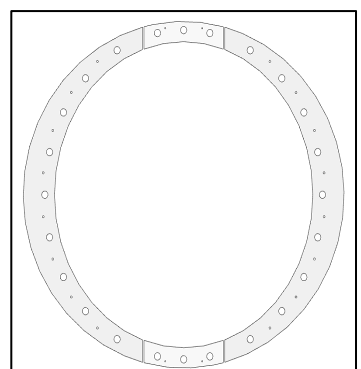


Fig. 7 York bracket

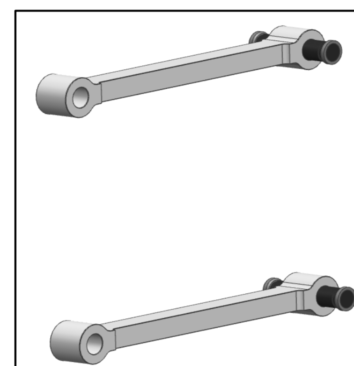


Fig. 8 Link & Link Pin

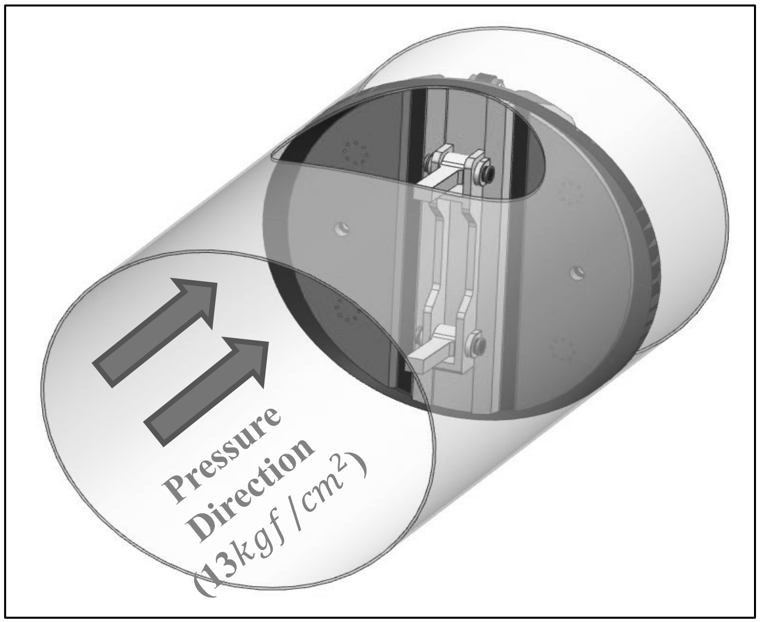
◆ 해석 조건

- York part
  - Material properties of York part components

Part	Material	Density (kg/m^3)	Young`s modulus(MPa)	Yield stress (MPa)	Tensile stress (MPa)	Poisson`s ratio
Element plate	A537 CL1	7820	210000	345	553	0.3
Link pin	S45C	7820	210000	490	686	0.3
Link	S45C	7820	210000	490	686	0.3
Yoke	A537 CL2	7820	210000	380	585	0.3
Element	RUBBER		70			0.49
York bracket	SS275	7820	210000	275	480	0.3

◆ 해석 조건

- York part
  - Boundary condition
    ① pressure : $15kgf/cm^2$($\approx$ 1.47MPa)
    ❖ Pressure direction is opposite to surface normal direction



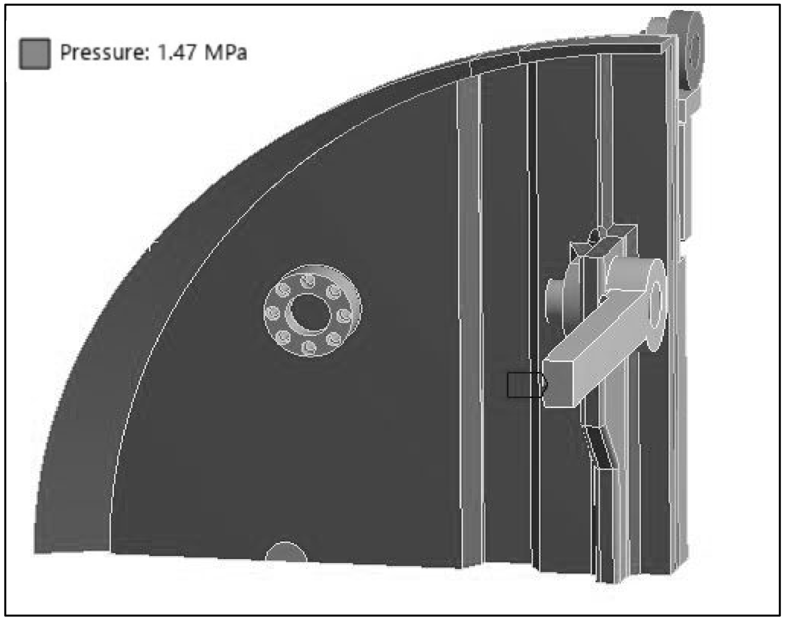


Fig. 9 Yoke part load condition

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◆ 해석 조건

- York part
  - Constraints
    ① Fixed support at the ends of Pipe surface for numerical singular problem
    ② Frictional conditions between Pipe and Elements

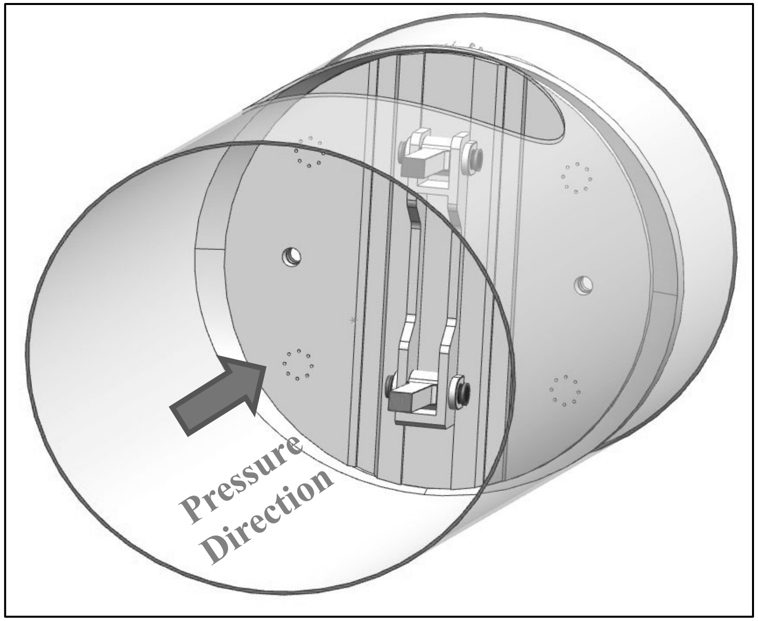


Fig. 10 constraint ① shown as red surfaces

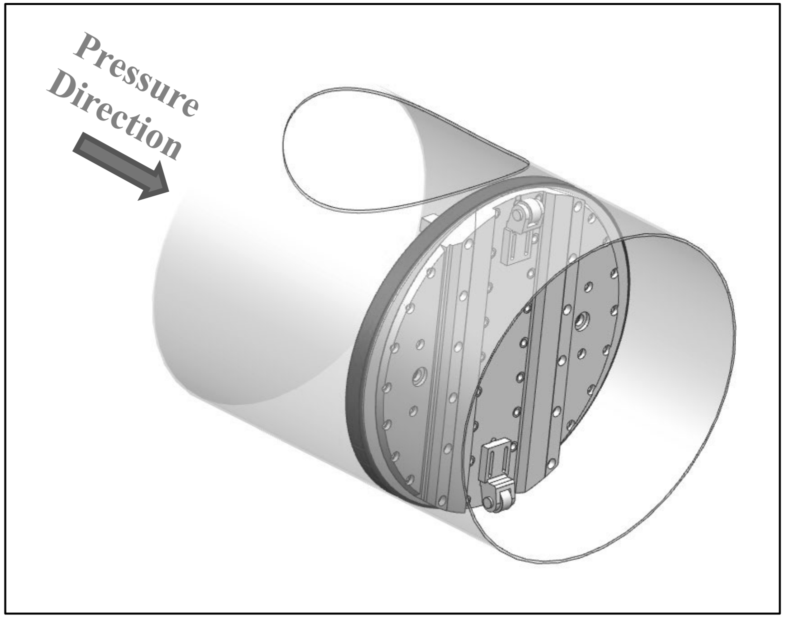


Fig. 11 constraint ② shown as red surfaces

◆ 해석 조건

- York part
  - Constraints
    - ③ Frictional conditions between each part surface
    - ④ Line bonded condition for combining each parts

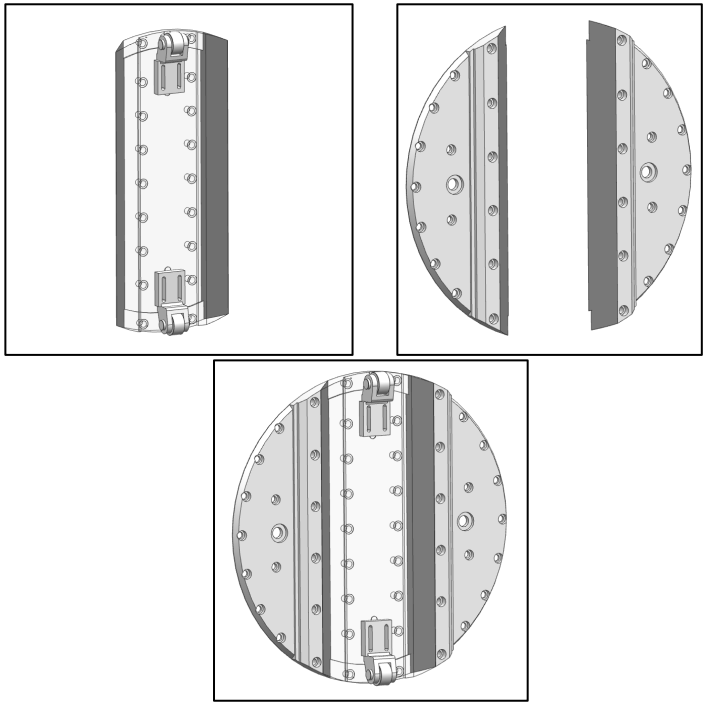


Fig. 12 constraint ③ shown as red and blue surfaces

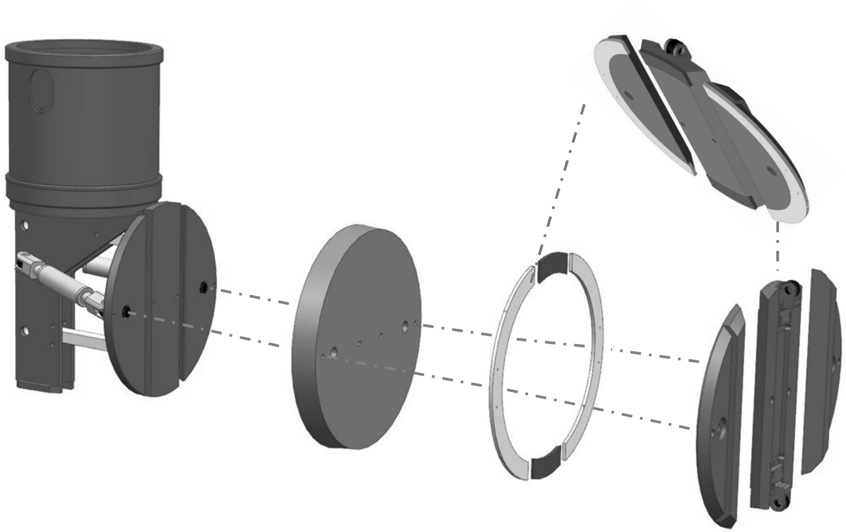


Fig. 13 constraint ④

◆ 해석 조건

- Push cylinder part
  - Components

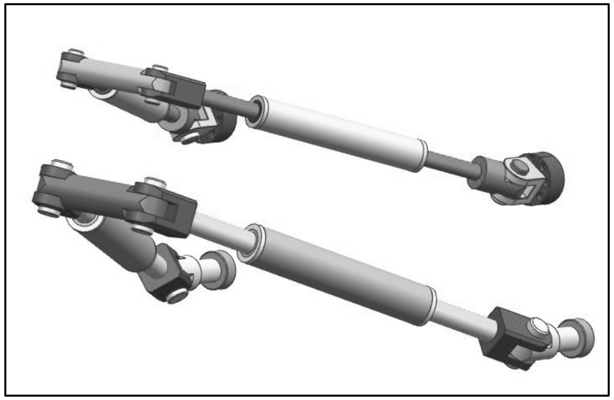


Fig. 14 Push part

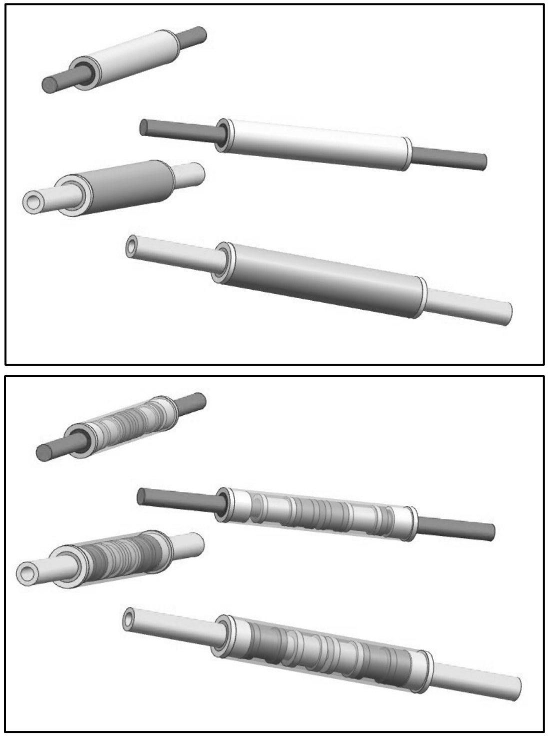


Fig. 15 Main push & Support Push

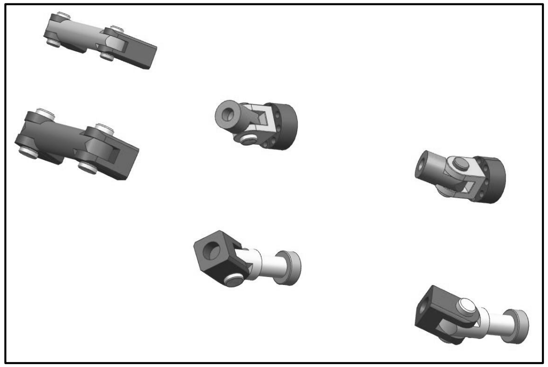


Fig. 16 Pin & Adapter & Cover

◆ 해석 조건

- Push cylinder part
  - Material properties of Push part components

Part	Material	Density (kg/m^3)	Young`s modulus(MPa)	Yield stress (MPa)	Tensile stress (MPa)	Poisson`s ratio
Push rod	S45C	7820	210000	490	686	0.3
Cover	S45C	7820	210000	490	686	0.3
Adapter	S45C	7820	210000	490	686	0.3
Pin	S45C	7820	210000	490	686	0.3

◆ 해석 조건

- Push cylinder part
  - Boundary condition
    - ① Displacement indicated as vectors obtained from York part results

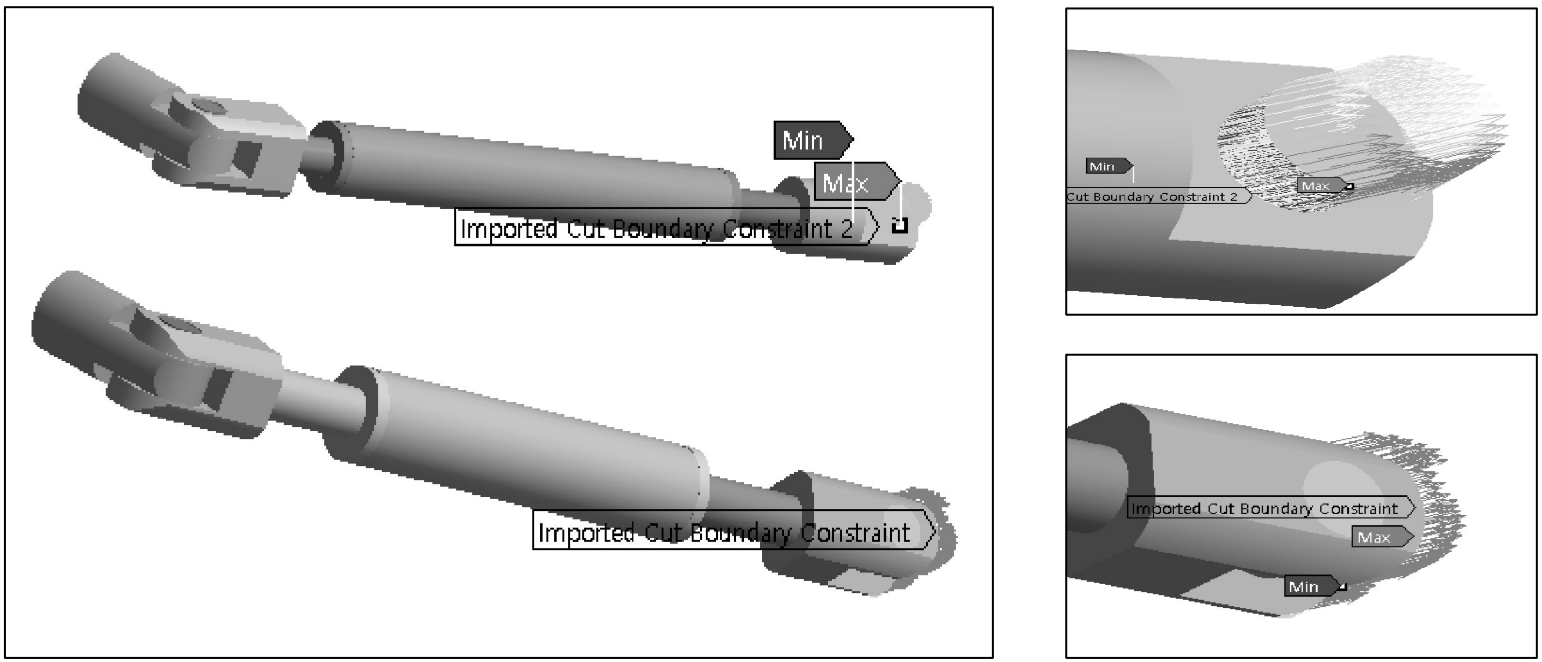


Fig. 17 Push part load condition

◆ 해석 조건

- Push cylinder part
  - Constraints
    ① Allowance for revolution
    ② Allowance for translation

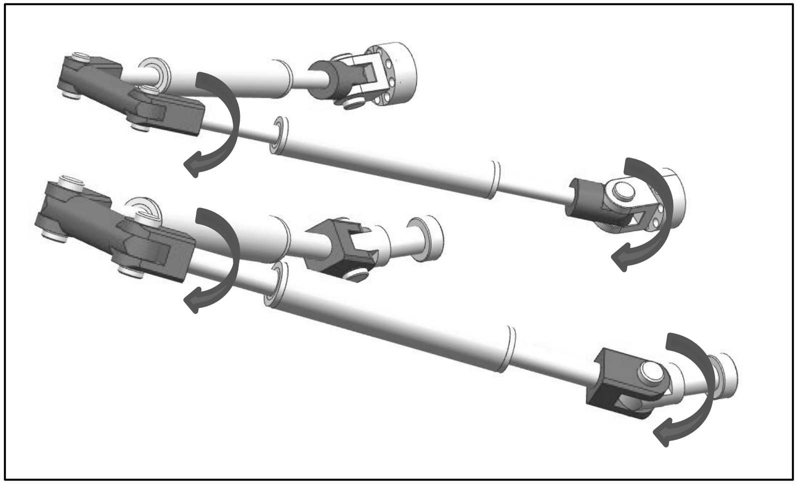


Fig. 18 constraint ① shown as red

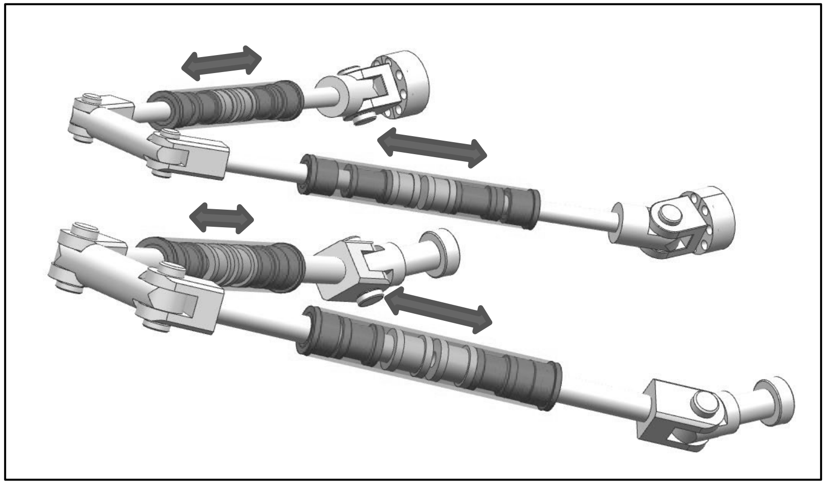


Fig. 19 constraint ② shown as red

◆ 해석 조건

- Guide support part
  - Components

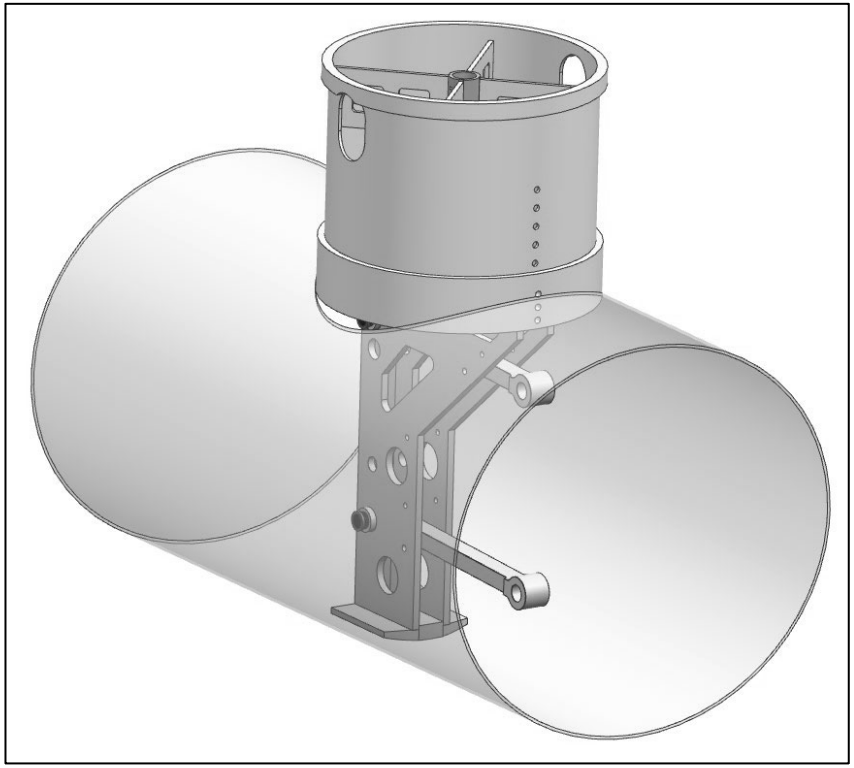


Fig. 20 Guide support part

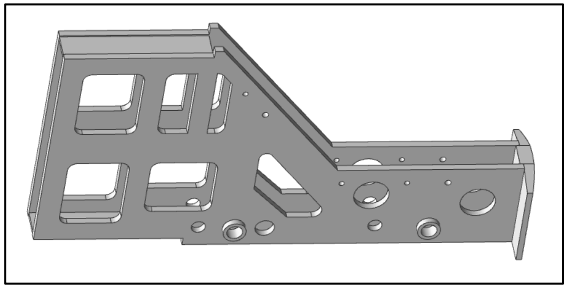


Fig. 21 Guide support

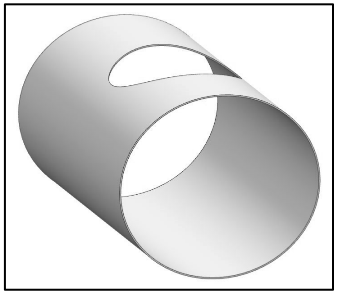


Fig. 23 Pipe

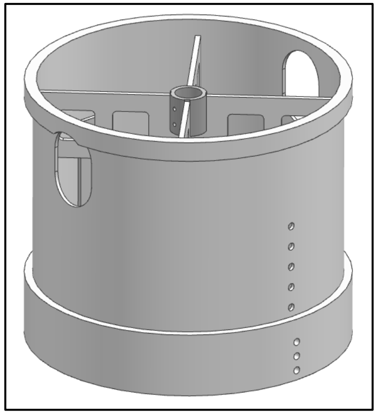


Fig. 22 Circular guide

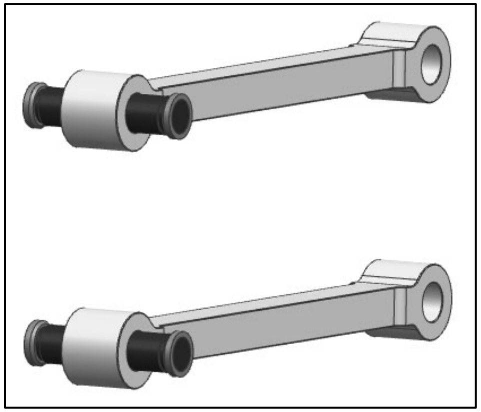


Fig. 24 Link & Pin

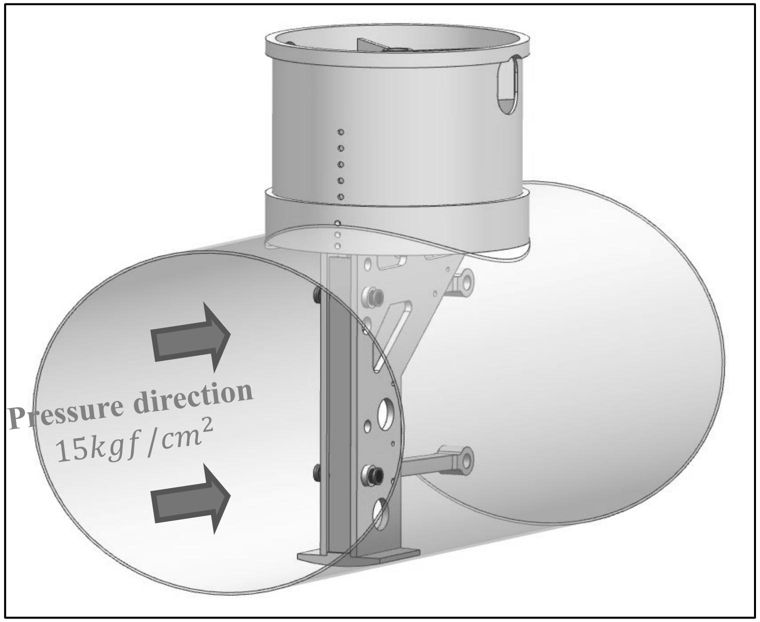
◆ 해석 조건

- Guide support part
  - Material properties of Guide support part components

Part	Material	Density (kg/m^3)	Young`s modulus (MPa)	Yield stress (MPa)	Tensile stress (MPa)	Poisson`s ratio
Guide support	SS275	7820	210000	275	480	0.3
Circular guide	SS275	7820	210000	275	480	0.3
Link	S45C	7820	210000	490	686	0.3
Pin	S45C	7820	210000	490	686	0.3

◆ 해석 조건

- Guide support part
  - Boundary condition
    - ① pressure : $15kgf/cm^2$($\approx$ 1.47MPa)
      - ❖ Pressure direction is opposite to surface normal direction



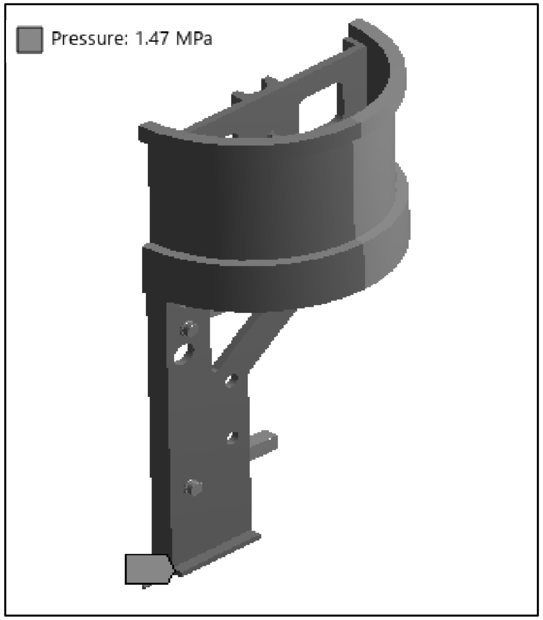
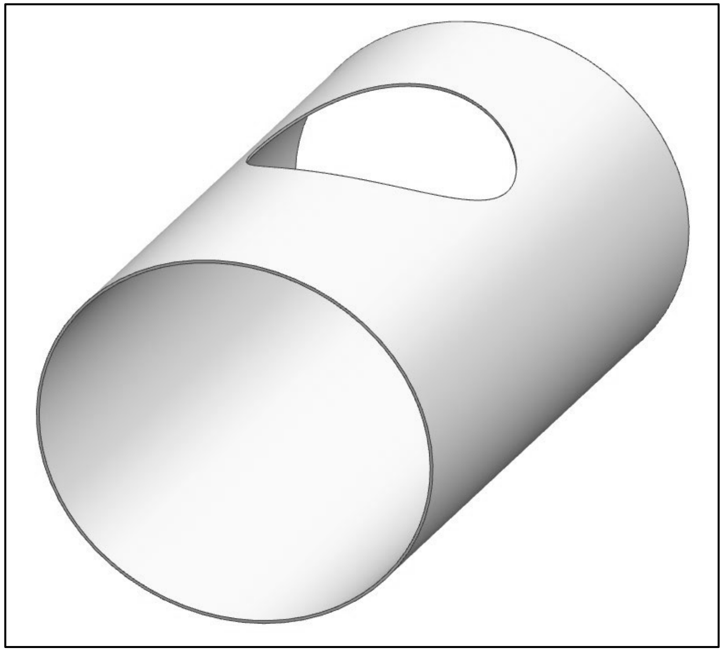


Fig. 25 Guide support part load condition

◆ 해석 조건

- Guide support part
  - Constraints
    - ① Fixed support at the ends of Pipe surface for numerical singular problem
    - ② Surface bonded condition for combining each parts



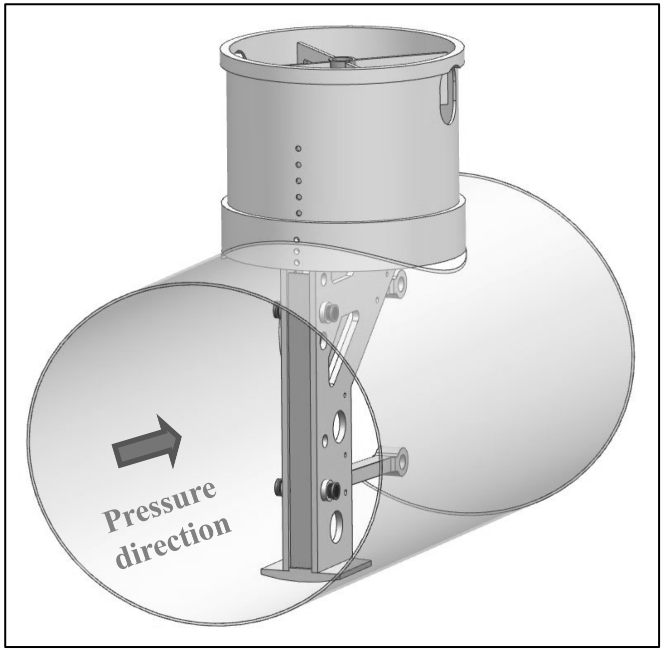


Fig. 26 constraint ① shown as red surfaces

◆ 해석 조건

- Methodology for Stress results

- Stress distribution, α

i : ith element of pipe surface
N : Total number of elements
σ_i : Stress on ith element(MPa)
A_i : Area of ith element(mm^2)
α_i : Stress distribution of ith element(%)

$$\alpha_i = \frac{\sigma_i A_i}{\sum_{k=1}^{N} \sigma_k A_k}$$

- Stress standard deviation, Z

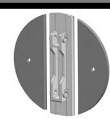
i : ith element of a component
N : Total number of a component
σ_i : Stress on ith element(MPa)
σ_{avg} : Average stress on a component (MPa)
S : Variance of a component
Z_i: Z score of ith element

$$S = \sqrt{\frac{\sum_{i=1}^{N} (\sigma_i - \sigma_{avg})^2}{N}}$$

$$Z_i = \frac{\sigma_i - \sigma_{avg}}{S}$$

◆ 1650A

- Yoke part

Part		Avg. Stress (MPa)	Max. Total Deformation (mm)
Element plate		38.18	27.22
Link and Pin		146.90	0.67
Element		1.30	27.71
Yoke		47.75	27.65
Yoke bracket		31.30	28.91

◆ 1650A

- Yoke part
  - Total deformation : 27.71 mm

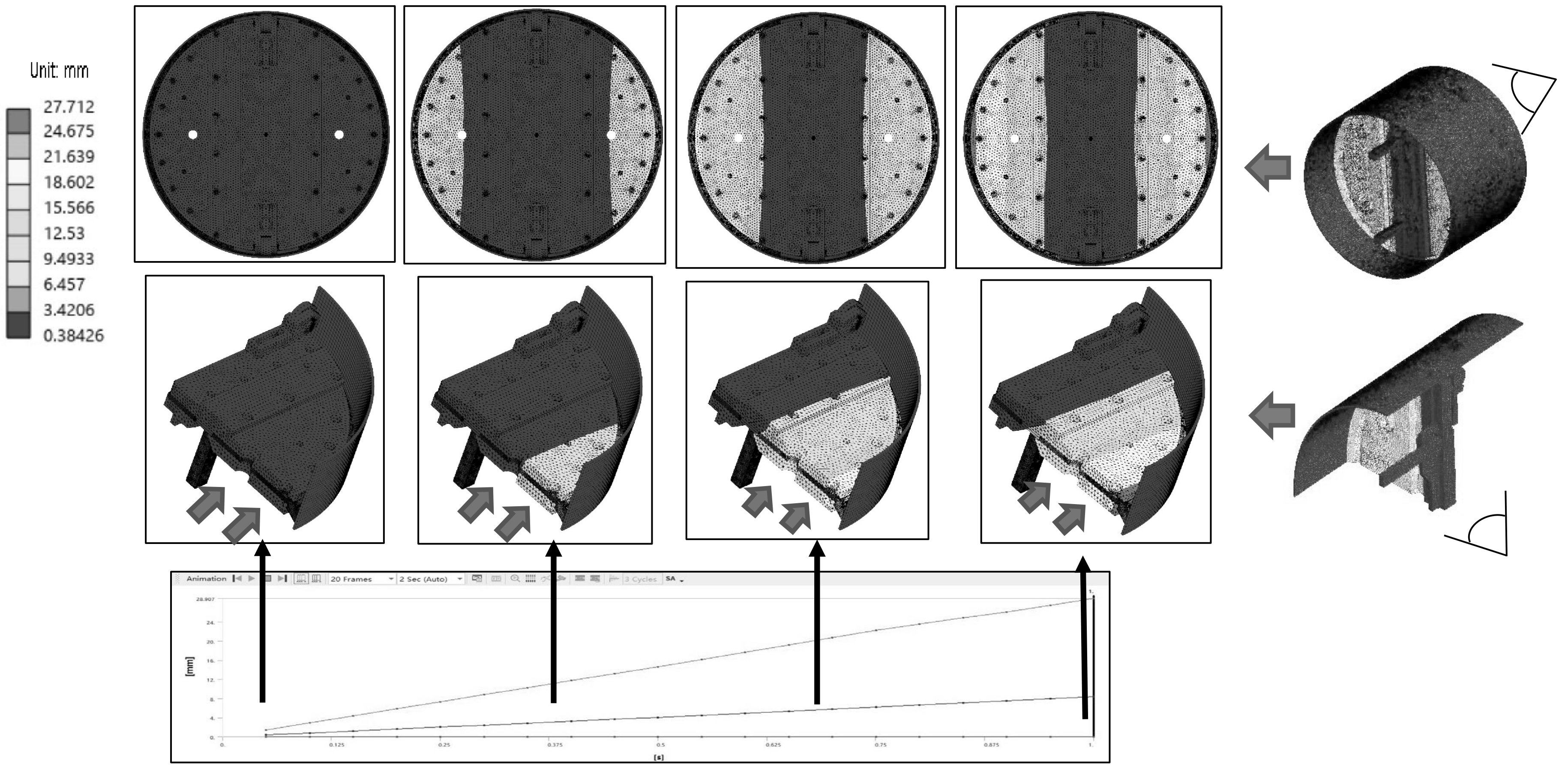


Fig. 27 Deformation results of Yoke part

◆ 1650A

- Yoke part
  - Element plate
    - ① Max. Total deformation : 27.22 mm
    - ② Avg. Stress : 38.18 MPa

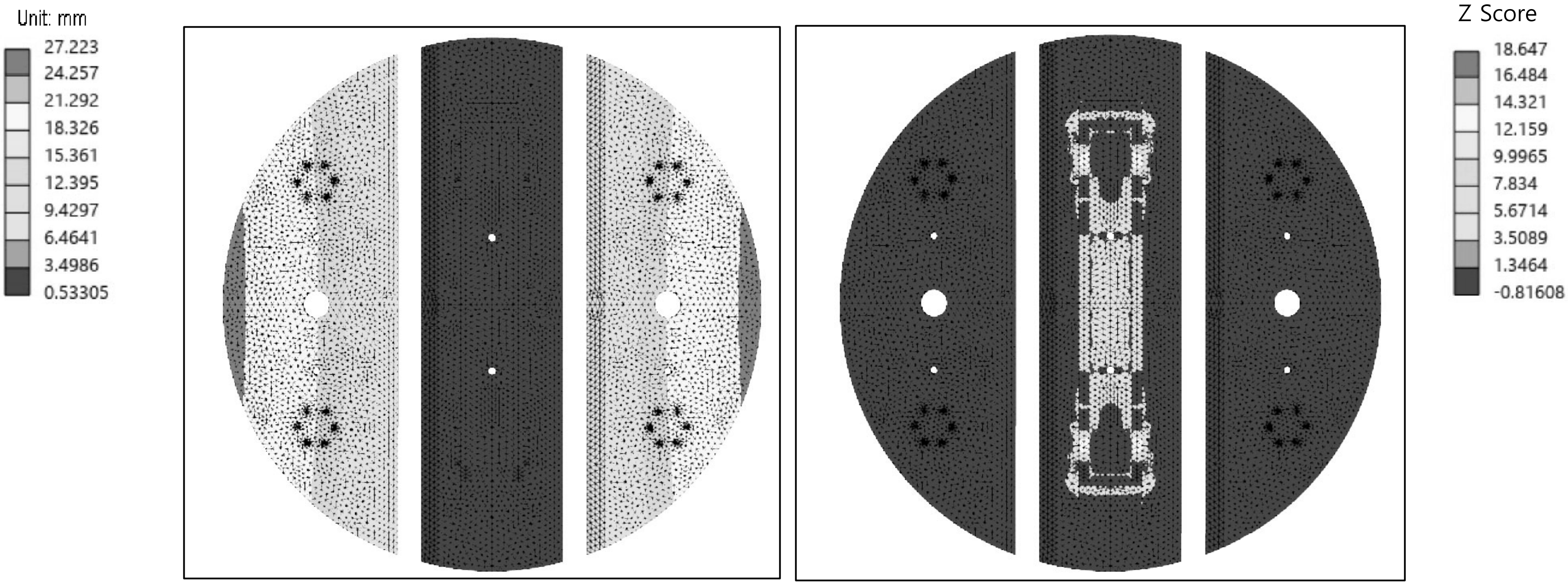


Fig. 29 Element plate results
(Left – Deformation / Right – Stress standard deviation)

◆ 1650A

- Yoke part
  - Link & Pin
    - ① Max. Total deformation : 0.67 mm
    - ② Avg. Stress : 146.90 MPa

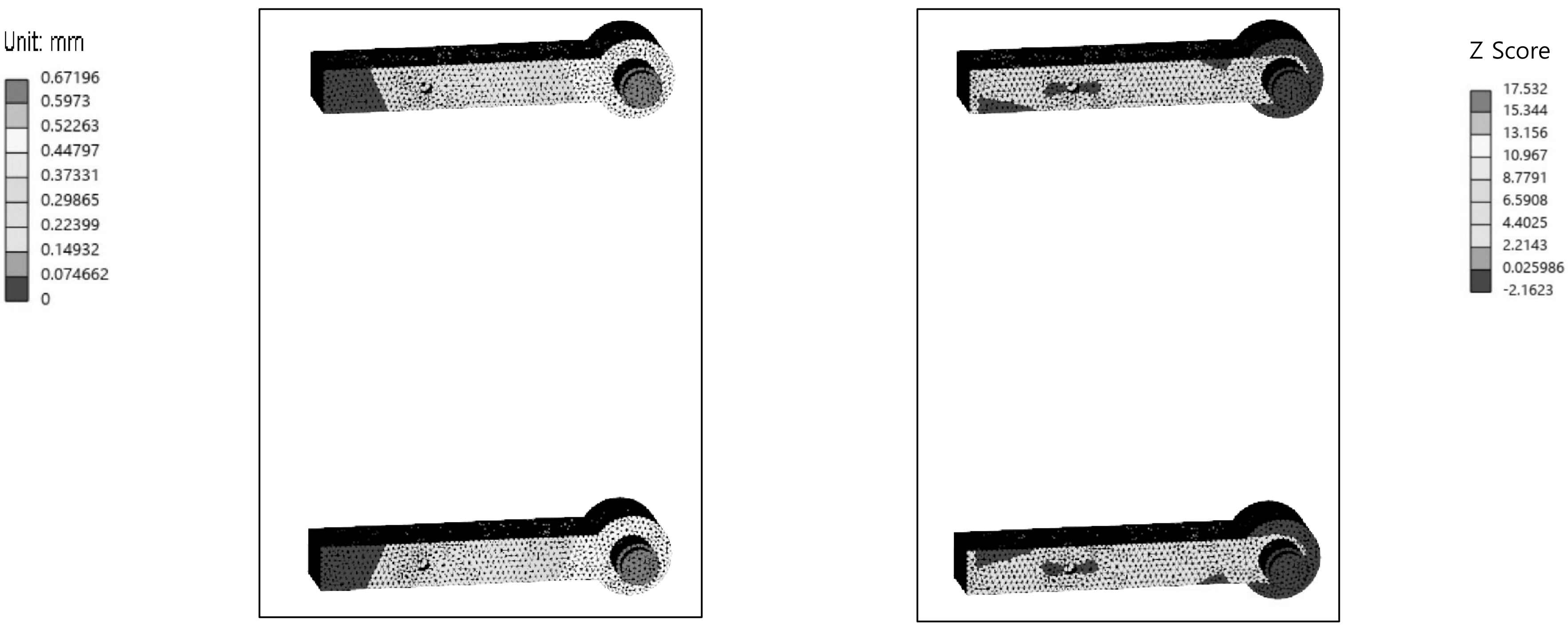


Fig. 30 Link & Pin results
(Left – Deformation / Right – Stress standard deviation)

◆ 1650A

- Yoke part
  - Element
    ① Max. Total deformation : 27.71mm
    ② Avg. Stress : 1.30MPa

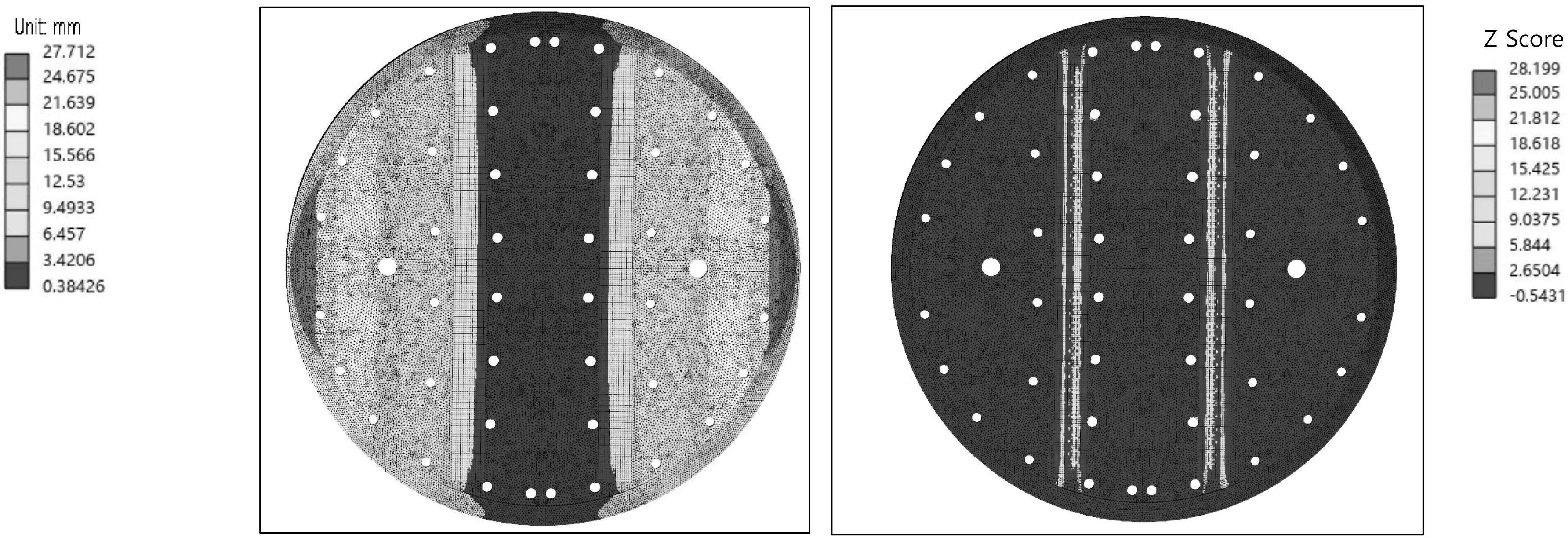


Fig. 31 Element results
(Left – Deformation / Right - Stress standard deviation)

◆ 1650A

- Yoke part
  - Yoke
    ① Max. Total deformation :27.65mm
    ② Avg. Stress : 47.75MPa

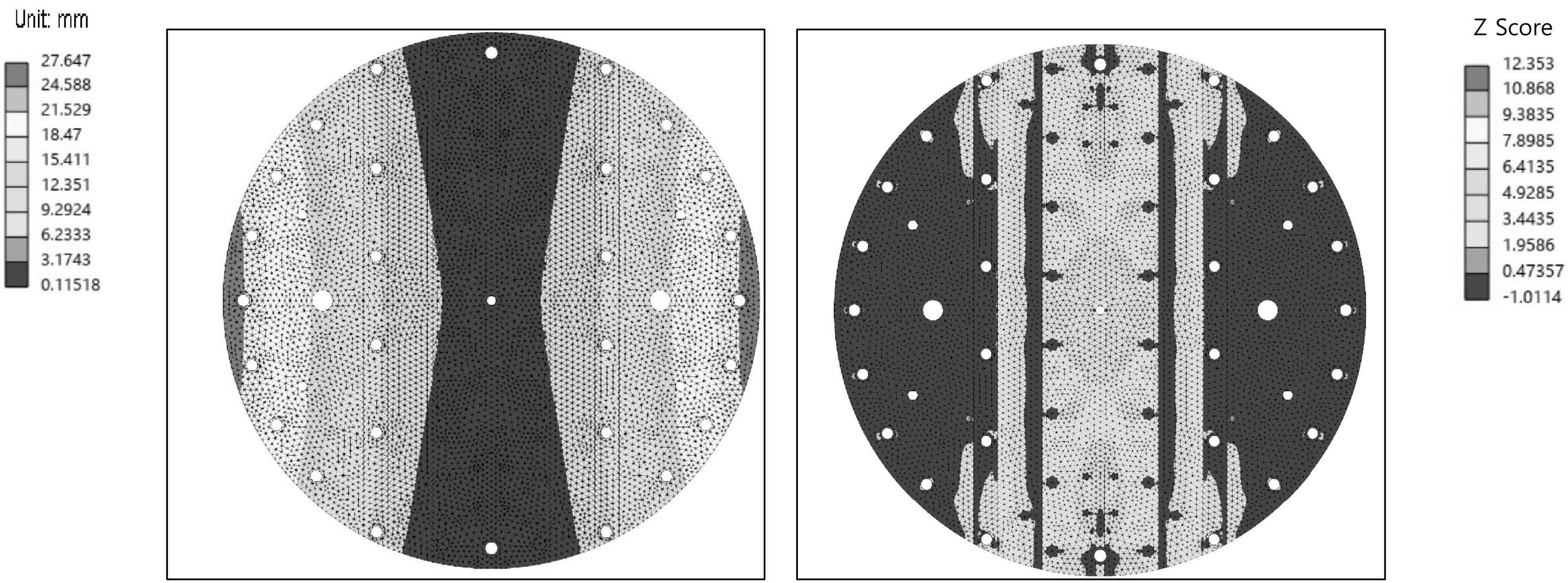


Fig. 32 York bracket results
(Left – Deformation / Right - Stress standard deviation)

◆ 1650A

- Yoke part
  - Yoke bracket
    - ① Max. Total deformation :28.91mm
    - ② Avg. Stress : 31.30MPa

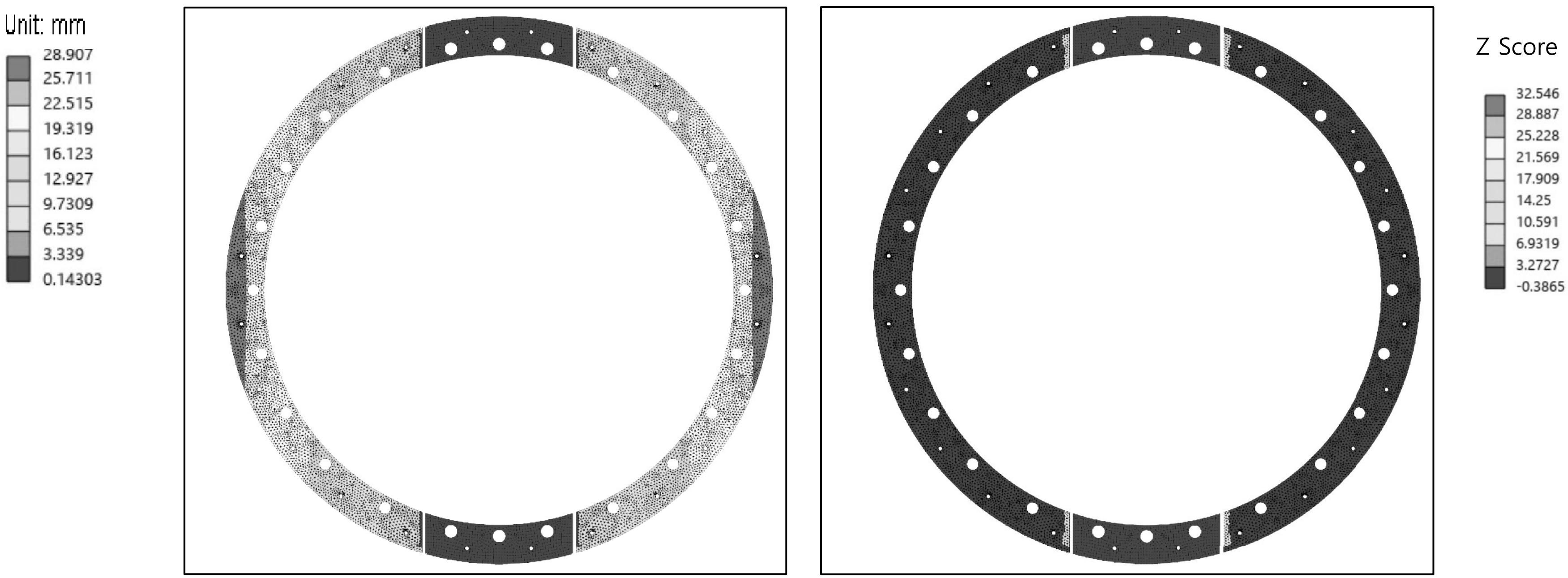
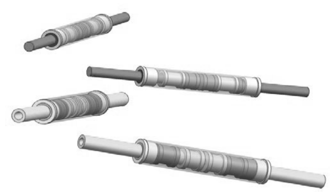


Fig. 33 York bracket results
(Left – Deformation / Right - Stress standard deviation)

◆ 1650A

- Push cylinder part

Part		Avg. Stress (MPa)	Max. Total Deformation (mm)
Push rod		1.53	17.24
Pin & Adapter		1.79	15.38

◆ 1650A

- Push cylinder part
  - Total deformation : 17.24 mm

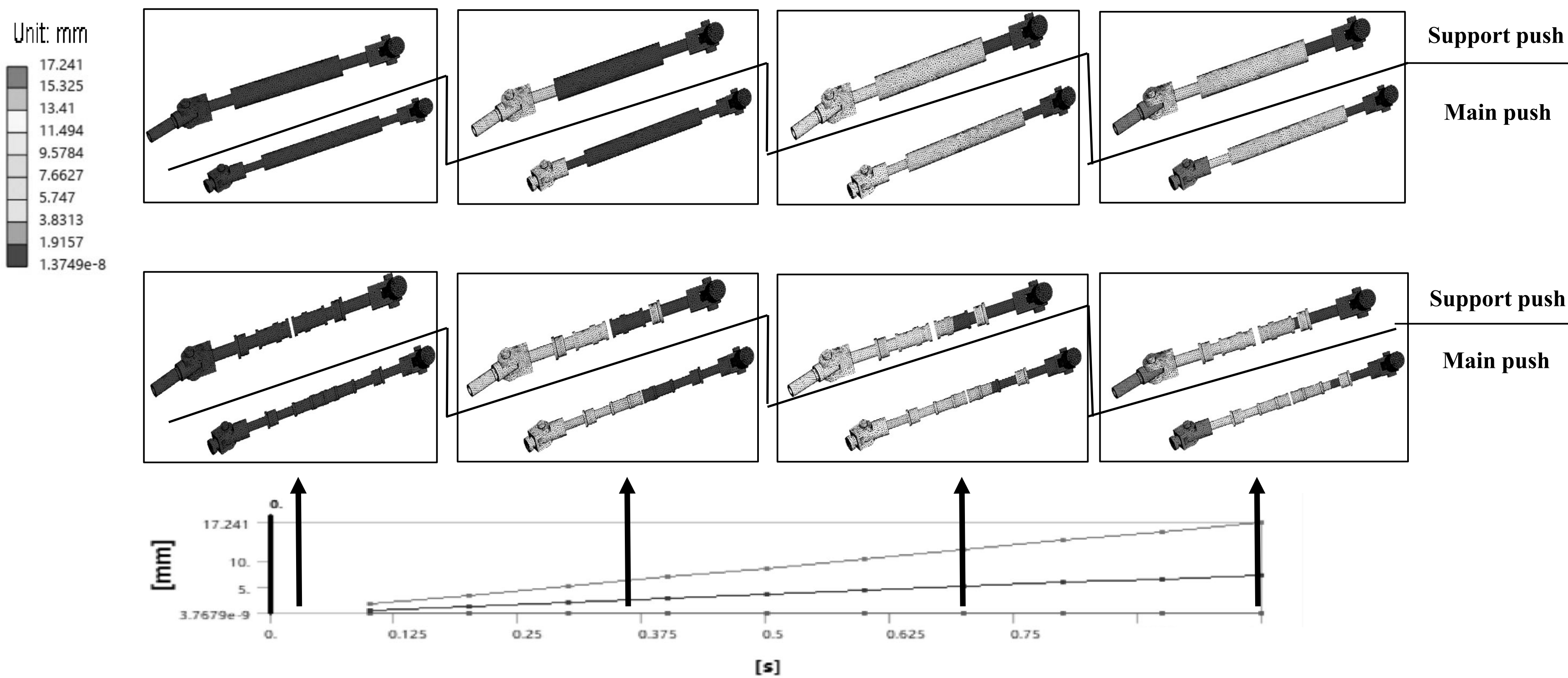


Fig. 34 Deformation results of Push part

◆ 1650A

- Push cylinder part
  - Total deformation : 17.24 mm

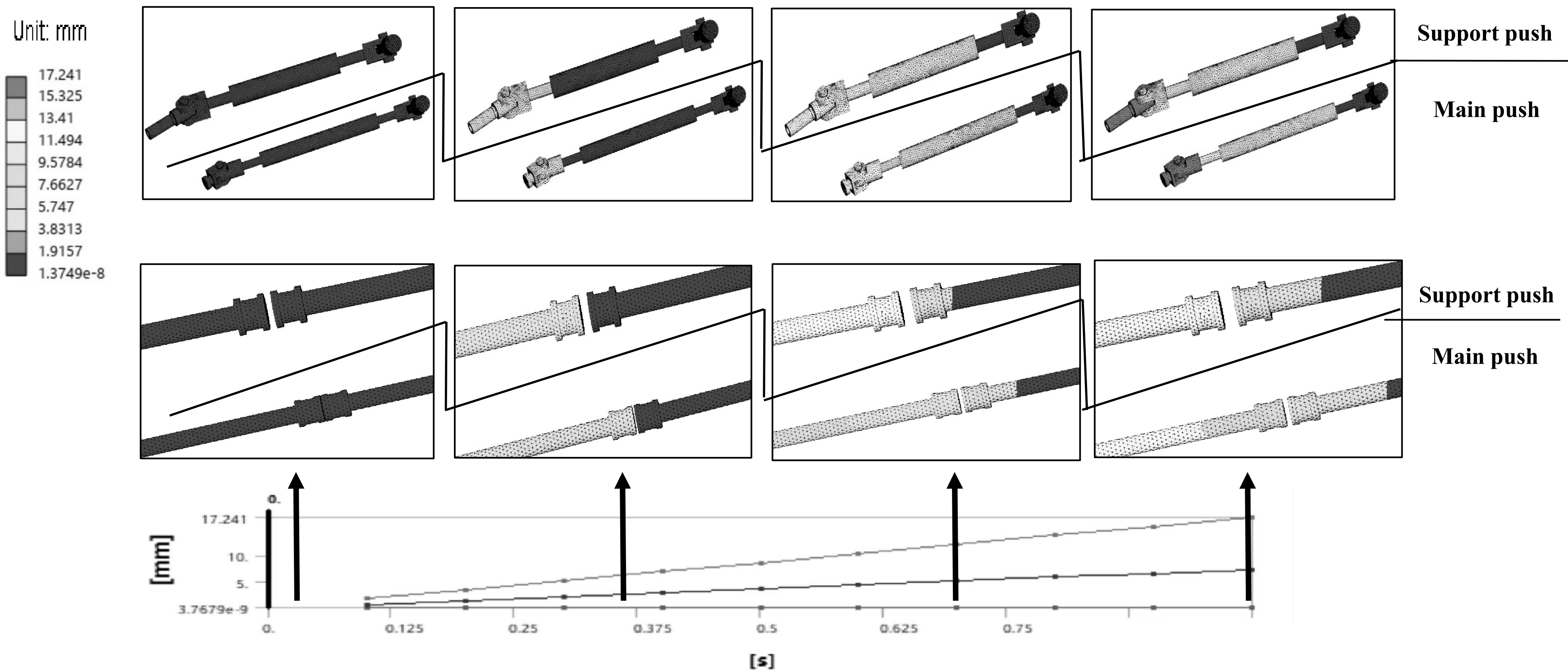


Fig. 34 Deformation results of Push part

1650A

- Push cylinder part
  - Pin & Adapter
    - ① Max. Total deformation :17.24mm
    - ② Avg. Stress :1.53MPa

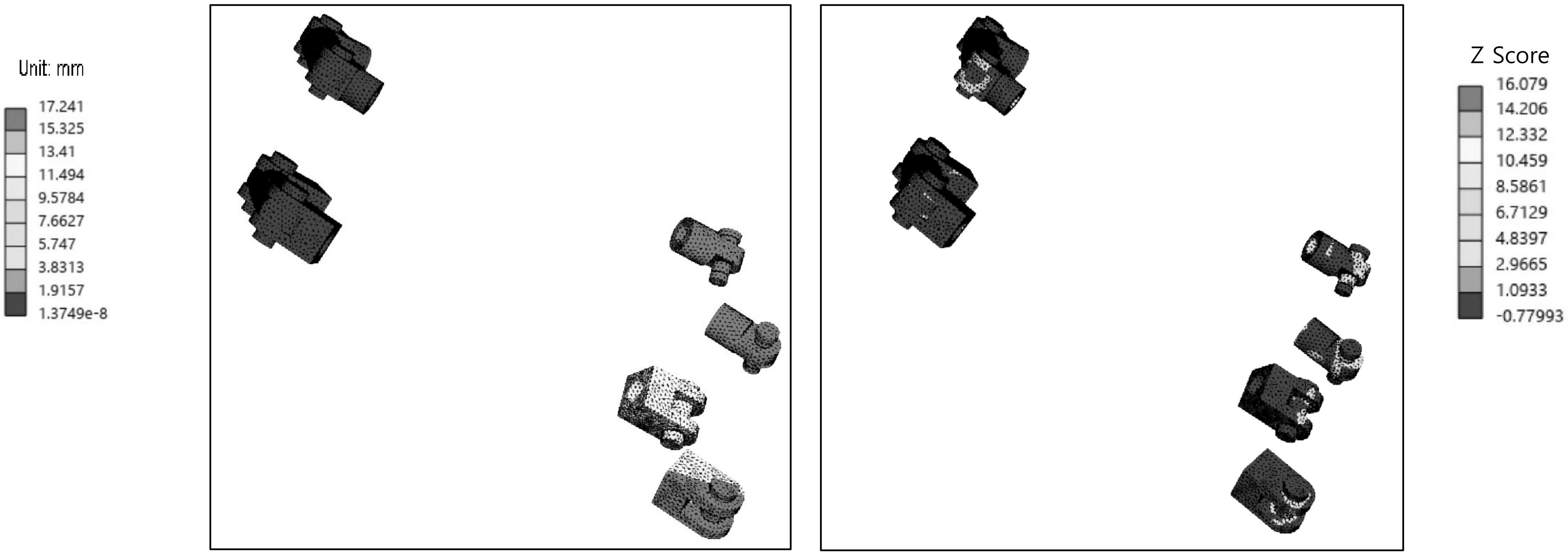


Fig. 35 Pin & Adapter results
(Left – Deformation / Right - Stress standard deviation)

◆ 1650A

- Push cylinder part
  - Push rods
    - ① Max. Total deformation :15.38mm
    - ② Avg. Stress :1.79MPa

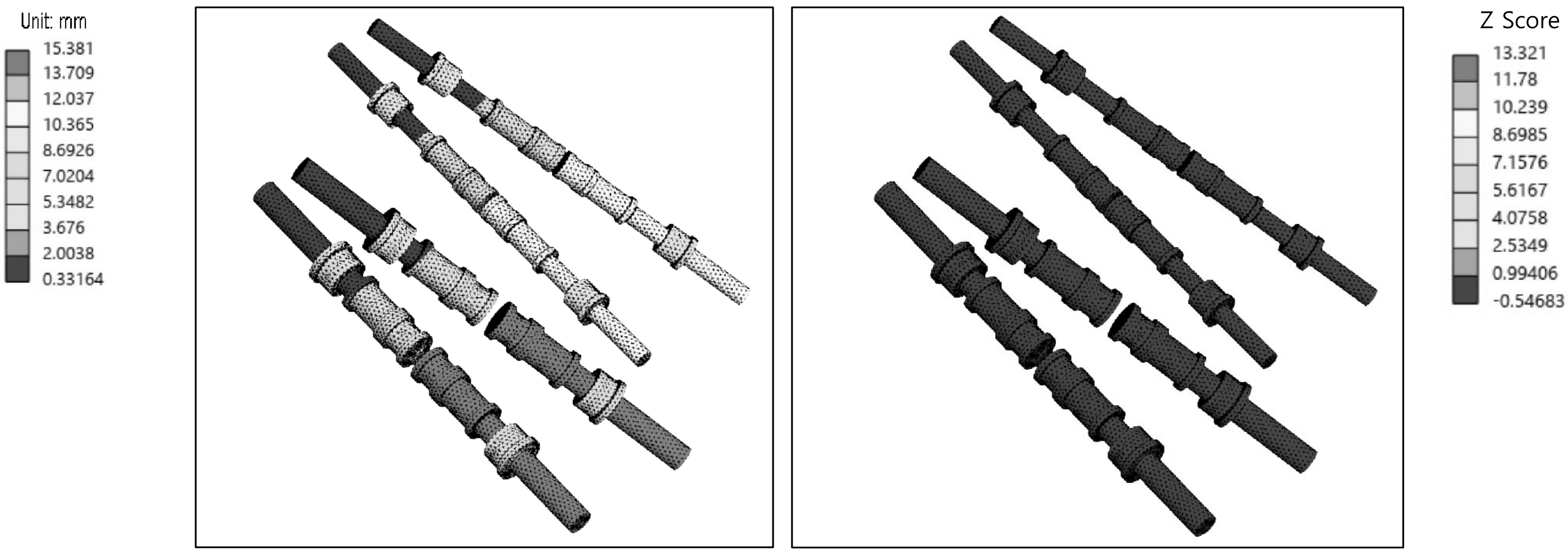
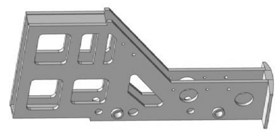
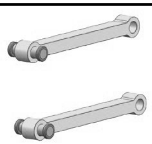


Fig. 36 Push rods results
(Left – Deformation / Right - Stress standard deviation)

1650A

- Guide support part

Part		Avg. Stress (MPa)	Max. Total Deformation (mm)
Guide support		90.90	9.74
Circular guide		20.11	0.62
Link & Pin		115.36	1.95

◆ 1650A

- Guide support part
  - Overall component
    - ① Max. Total deformation :9.74mm
    - ② Avg. Stress :38.48MPa

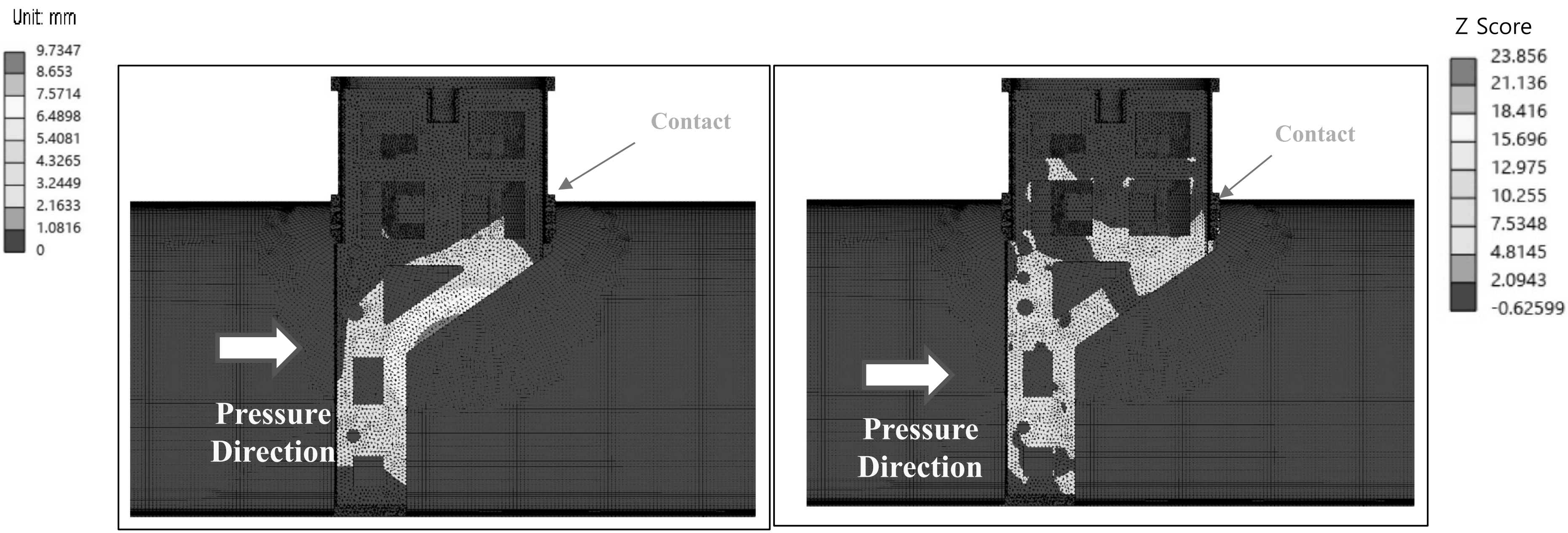


Fig. 37 Overall components results
(Left – Deformation / Right - Stress standard deviation)

◆ 1650A

- Guide support part
  - Guide support
    - ① Max. Total deformation :9.74mm
    - ② Avg. Stress : 90.90MPa

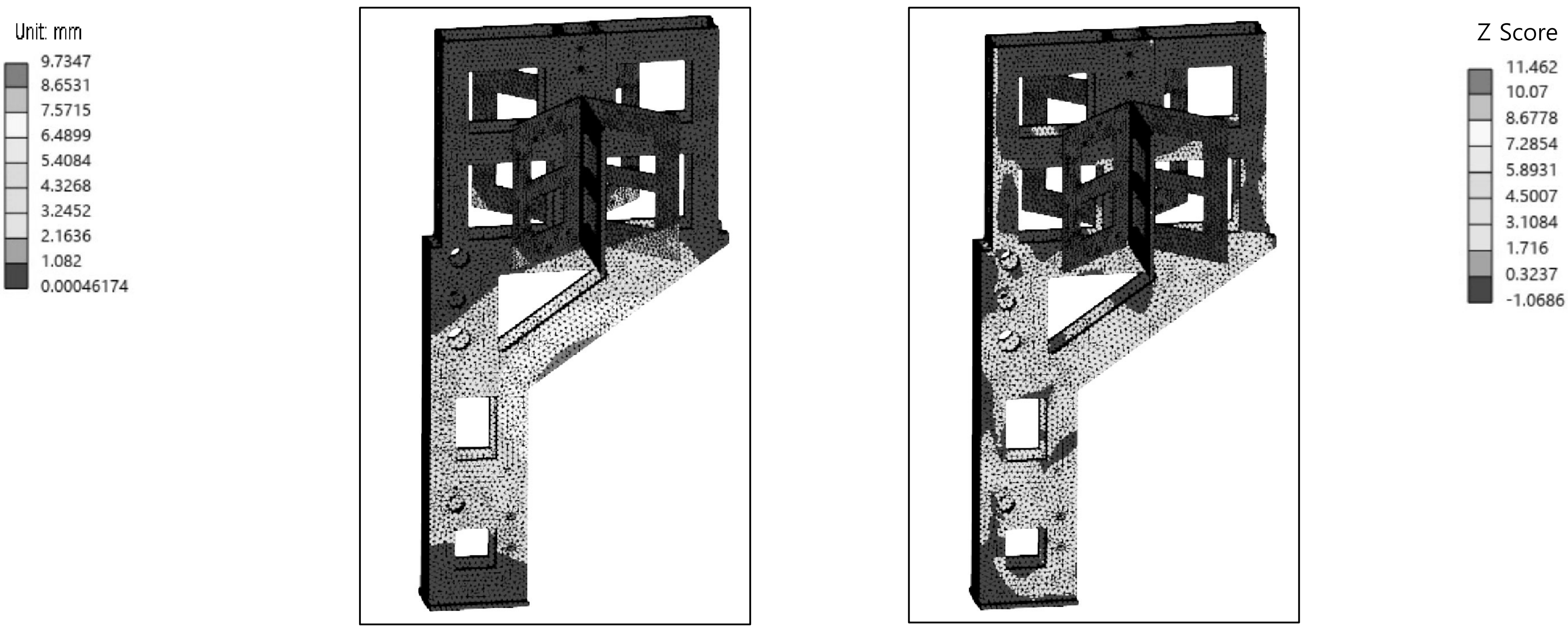


Fig. 38 Guide support results
(Left – Deformation / Right - Stress standard deviation)

◆ 1650A

- Guide support part
  - Circular guide
    ① Max. Total deformation :0.62mm
    ② Avg. Stress :20.11MPa

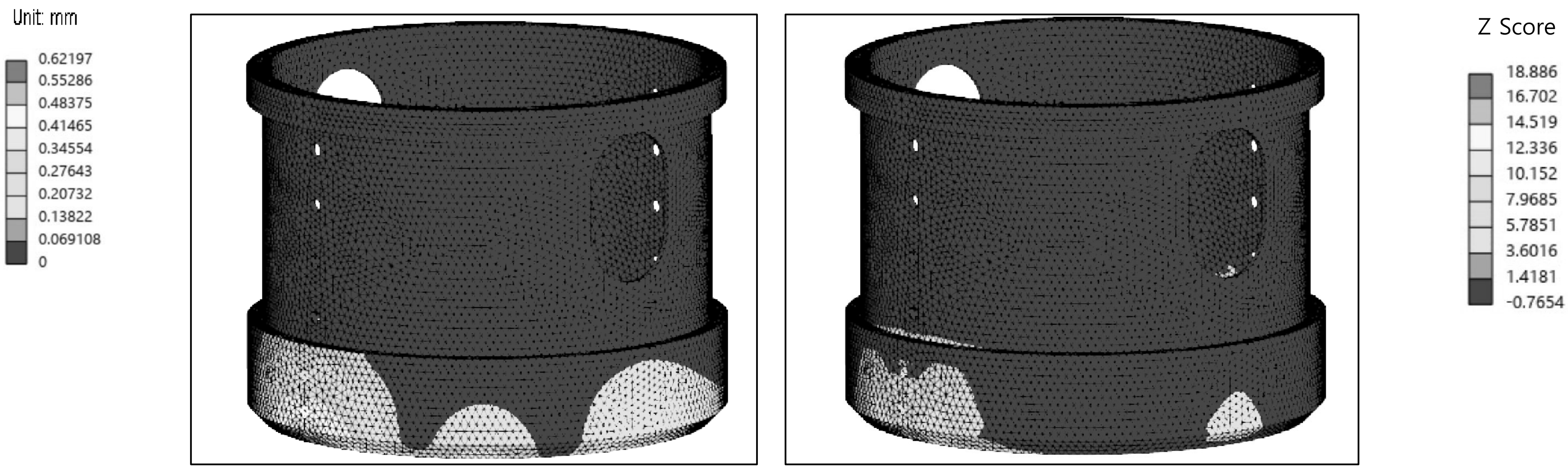


Fig. 39 Circular guide results
(Left – Deformation / Right - Stress standard deviation)

◆ 1650A

- Guide support part
  - Link and Pin
    ① Max. Total deformation :1.95mm
    ② Avg. Stress :115.36MPa

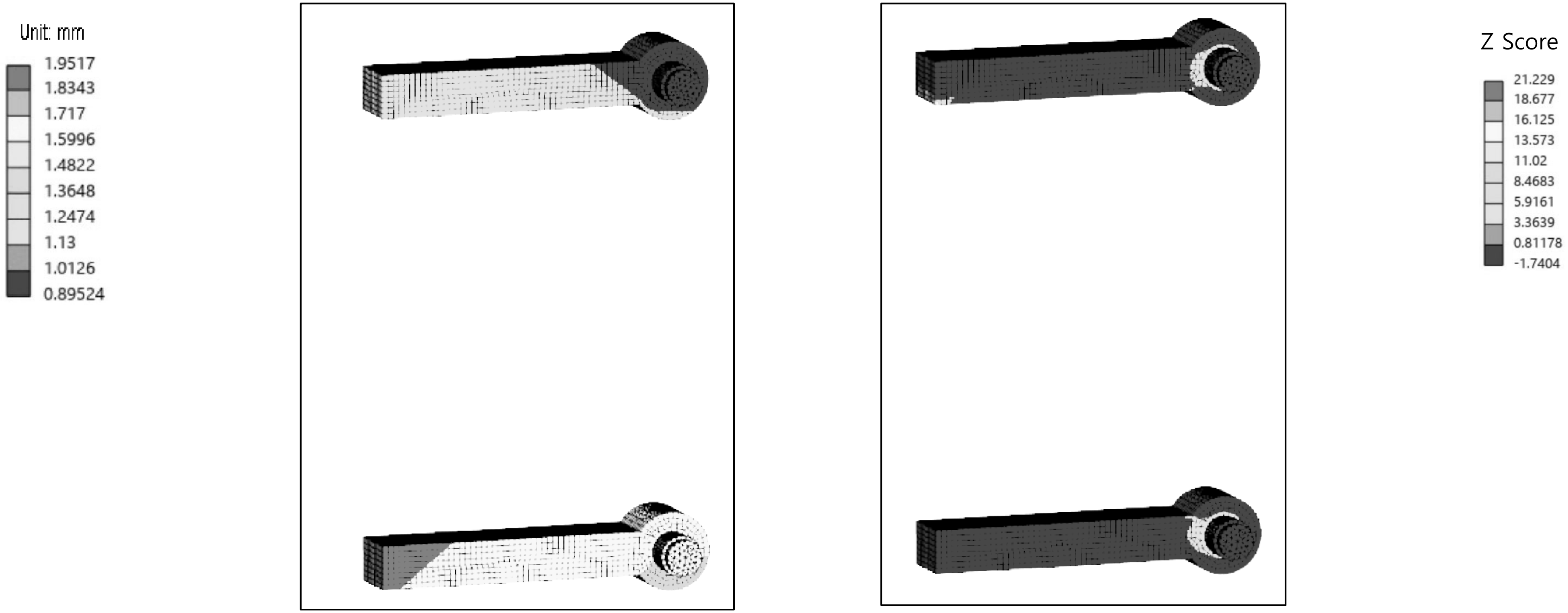


Fig. 40 Link and Pin results
(Left – Deformation / Right - Stress standard deviation)

◆ 구조해석을 통한 1650A 배관 차단장치의 안정성 분석

- 구조해석 결과, 안정성 평가에 대한 결론은 다음과 같다.

① 각 요소의 평균 응력은 재료의 항복강도 이하로 계산됨.

② Yoke part의 경우, $15kgf/cm^2$ 압력 조건에서 유체 차단이 가능함을 확인함.

- Yoke part가 이상적인 범위 내에서 변형되고 응력 분포가 균일하다
- Rubber와 배관 내벽은 장치 작동 중에 밀착되어 효과적인 밀봉을 보장함

③ Push cylinder part의 경우, $15kgf/cm^2$ 압력 조건에서도 Push cylinder 사이의 간극이 존재하는 것을 확인함.

- Push cylinder 내부의 모든 부품에 간섭이나 과도한 변형이 없음
- 안전 여유가 높고 반복 운전 시 안정성 확보함

④ Guide support part의 경우， $15kgf/cm^2$ 압력 조건에서 Yoke part와 Push cylinder part를 안정적으로 지지하여 배관 차단장치의 정상적인 운전을 확보함

- Guide support part내부의 각 부품에 경사와 오정렬이 나타나지 않아 효과적인 지지를 제공함